FinMarket-Enterprise

Phase II — Database and Security Integration

Member 3: @KvngWilson — Marketing Campaigns and Templates

Branch: `feature/campaign-p2-m3`
Assigned module: `backend/src/modules/campaigns/`
Primary ownership: Campaign and template persistence, campaign lifecycle management, audience selection, advisor ownership, specialist campaign assignments, and campaign authorization.

---

1. Project Overview

FinMarket-Enterprise is a financial advisory marketing and client acquisition platform designed to help financial advisors manage prospective clients, organize marketing campaigns, deliver authorized marketing communications, automate follow-up activities, and monitor marketing performance.

The application consists of five main modules:

1. Identity and Advisor Management
2. Lead Management (CRM)
3. Marketing Campaigns and Templates
4. Email and Facebook Delivery
5. Automation and Analytics

During Phase I, the Marketing Campaigns module was implemented using temporary in-memory storage. Core APIs were established for campaign creation, retrieval, updating, template management, and audience selection.

Initial Mongoose models, campaign strategies, lifecycle statuses, and validation rules were also introduced.

Phase II focuses on replacing temporary campaign and template storage with MongoDB, enforcing secure advisor ownership, implementing campaign-specific

Marketing Specialist assignments, and maintaining the relationships between campaigns, templates, leads, and delivery operations.

The existing Campaigns architecture and API contracts should be preserved wherever practical.

1.1 Phase II Objectives

The following responsibilities must be completed:

- Replace in-memory Campaign and Template storage with MongoDB using Mongoose.
- Implement persistent Campaign and Template records.
- Maintain relationships between Campaigns, AdvisorProfiles, Templates, and Leads.
- Protect campaign and template endpoints using JWT authentication.
- Enforce role-based authorization and resource ownership.
- Implement campaign-specific Marketing Specialist assignments.
- Ensure specialists can access only campaigns explicitly assigned to them.
- Enforce campaign lifecycle and status transitions.
- Validate campaign activation requirements.
- Implement secure audience selection using eligible Lead records.
- Prevent unauthorized campaign modifications.
- Maintain compatibility with Delivery and Automation.
- Preserve existing Phase I API functionality.
- Implement Postman and automated tests for persistence, validation, and authorization.

The Campaigns module must remain independently developable and testable without requiring other members to complete their feature implementations.

---

2. Development Environment and Team Independence

The Dockerized backend and MongoDB infrastructure will be prepared by Member 1 and merged into `main` before Phase II feature development begins.

Once the infrastructure is available, the repository must be cloned or updated from the latest `main` branch.

Development must occur in:

`feature/campaign-p2-m3`

The shared Docker configuration, MongoDB connection, authentication middleware, role constants, and error-handling contracts must be reused.

Member 3 must not independently replace or modify shared infrastructure.

The Campaigns module must remain responsible for its own:

- Mongoose models.
- MongoDB repositories.
- Campaign services and business rules.
- Controllers and routes.
- Campaign and template validation.
- Campaign-specific specialist assignments.
- Audience selection.
- Module-specific authorization.
- Unit and integration tests.
- Module documentation.

Changes to shared files such as `backend/src/app.js`, `backend/src/routes/index.js`, `backend/src/config/`, or authentication middleware must be coordinated with Member 1.

2.1 Independent Development Requirements

The following approach must be followed:

1. Existing Campaign and Template API contracts must be preserved wherever practical.
2. The in-memory repository must be replaced without requiring changes to other members' modules.
3. Stable MongoDB ObjectIds must be used to represent relationships.
4. Sample AdvisorProfiles, Users, Leads, and accepted specialist assignments may be created directly through test fixtures.
5. Shared authentication and relationship lookup contracts may be mocked during isolated tests.

6. The Campaigns module must not require the Delivery or Automation modules to operate during independent development.
7. Audience selection must use an agreed Lead lookup contract rather than modifying the Leads module.
8. Specialist authorization must use the agreed Identity assignment contract rather than implementing a second invitation system.

The completed module must integrate with the shared security infrastructure and MongoDB connection.

---

3. Shared User Roles and Permissions

The application supports three roles:

3.1 Administrator — `ADMIN`

Administrators manage platform-level user accounts, permissions, and administrative operations.

Administrative access to Campaigns must follow explicitly defined permissions.

The Administrator role must not automatically bypass all resource ownership checks unless such access has been intentionally granted by the shared authorization policy.

3.2 Financial Advisor — `ADVISOR`

Financial Advisors own campaigns and the associated marketing activities.

An authenticated Advisor must be allowed to:

- Create campaigns under their own AdvisorProfile.
- View their own campaigns.
- Update their own campaigns.
- Create and manage their own marketing templates.
- Define target audiences.
- Activate, schedule, complete, or cancel campaigns according to lifecycle rules.
- Assign accepted Marketing Specialists to their campaigns.
- Remove specialists from their campaigns.
- View the specialists assigned to their campaigns.

An Advisor must not access or modify another Advisor's campaigns without explicitly authorized access.

3.3 Marketing Specialist — `MARKETING_SPECIALIST`

Marketing Specialists perform authorized marketing activities on behalf of Financial Advisors.

A Marketing Specialist may:

- View campaigns specifically assigned to them.
- Edit permitted content belonging to assigned campaigns.
- Prepare campaign content and templates when authorized.
- View eligible audience summaries where permitted.
- Access campaign information required for authorized delivery activities.

A Marketing Specialist must not:

- Access unrelated campaigns.
- Assign themselves to campaigns.
- Invite or assign other Marketing Specialists on behalf of Advisors.
- Change campaign ownership.
- Access unrestricted Lead records.
- Modify AdvisorProfile information.
- Perform administrative operations.

Two conditions must be satisfied before a specialist receives campaign access:

1. An accepted advisor-specialist relationship must exist.
2. The specialist must be explicitly assigned to the campaign.

An accepted invitation alone must not grant access to every campaign belonging to the Advisor.

The backend must enforce both conditions.

---

4. Database Identifiers and Relationships

The following identifier conventions must be used consistently.

4.1 Identifier Definitions

Field	Meaning	Referenced Collection
`userId`	Unique identifier of a registered user	`users`
`advisorId`	Unique identifier of an AdvisorProfile	`advisorProfiles`
`specialistUserId`	User ID belonging to a Marketing Specialist	`users`
`campaignId`	Unique identifier of a Campaign	`campaigns`
`templateId`	Unique identifier of a Template	`templates`
`leadId`	Unique identifier of a Lead	`leads`
`deliveryJobId`	Unique identifier of a DeliveryJob	`deliveryJobs`
`ownerId`	User ID of the owner where defined	`users`

Important distinction:

`advisorId` always references `AdvisorProfile._id`, not `User._id`.

For example:

- `User._id = A`
- `AdvisorProfile._id = B`
- `AdvisorProfile.userId = A`
- `Campaign.advisorId = B`
- `Template.advisorId = B`

For a Marketing Specialist:

- `User._id = C`
- `Campaign.assignedSpecialistIds = [C]`

Marketing Specialists do not have AdvisorProfile records.

Therefore, `assignedSpecialistIds` must reference the User collection.

4.2 Required Relationships

AdvisorProfile → Campaign

Relationship: One-to-many.

One AdvisorProfile may own multiple campaigns.

Each Campaign must belong to exactly one AdvisorProfile.

`Campaign.advisorId → AdvisorProfile._id`

AdvisorProfile → Template

Relationship: One-to-many.

Each Template belongs to an AdvisorProfile.

`Template.advisorId → AdvisorProfile._id`

Templates belonging to one Advisor must not automatically be accessible to another Advisor.

Template → Campaign

Relationship: One-to-many.

A Template may be reused by multiple campaigns.

A Campaign may reference one selected Template through:

`Campaign.templateId → Template._id`

The selected Template must belong to the same Advisor and be compatible with the Campaign channel.

Campaign → Lead

Relationship: Audience-based, potentially many-to-many.

A Campaign may target multiple eligible Leads.

A Lead may qualify for multiple Campaigns.

Audience membership must be determined using Lead attributes and Campaign targeting criteria.

A separate campaign-to-lead collection is not required for Phase II.

Campaign → Marketing Specialist

Relationship: Many-to-many.

A Campaign may have multiple assigned Marketing Specialists.

A Marketing Specialist may be assigned to multiple Campaigns.

The Campaign model must maintain:

`assignedSpecialistIds → User._id[]`

Each referenced User must have the `MARKETING_SPECIALIST` role.

The specialist must also have an accepted relationship with the Campaign's owning Advisor.

Campaign → DeliveryJob

Relationship: One-to-many.

A Campaign may produce multiple DeliveryJobs.

`DeliveryJob.campaignId → Campaign._id`

The Campaigns module must provide reliable campaign lookup and status information to the Delivery module.

The Delivery module remains responsible for delivery processing.

---

5. Member 3 — Assigned Phase II Responsibilities

5.1 Campaign Mongoose Model and Persistence

Owned directory: `backend/src/modules/campaigns/`

The existing Campaign model must be reviewed and updated for persistent MongoDB storage.

Collection:

`campaigns`

Recommended fields include:

- `_id`
- `advisorId`
- `title`
- `description`
- `strategy`
- `channel`
- `audience`
- `templateId`
- `assignedSpecialistIds`
- `status`
- `scheduledAt`
- `completedAt`
- `createdAt`
- `updatedAt`

The existing Phase I model fields and enum values must be preserved unless changes are necessary and documented.

Requirements include:

1. Required field validation.
2. Correct Mongoose field types.
3. Allowed campaign strategies.
4. Allowed marketing channels.
5. Allowed lifecycle statuses.
6. Appropriate default values.
7. Timestamp management.
8. AdvisorProfile references.
9. Template references.

10. Marketing Specialist references.
11. Appropriate database indexes.

The Campaign model must not use the User ID as its `advisorId`.

Advisor ownership must be represented through `AdvisorProfile._id`.

The `assignedSpecialistIds` field should default to an empty array.

Duplicate specialist IDs must not be stored within the same campaign assignment list.

5.2 Template Mongoose Model and Persistence

Collection:

`templates`

Recommended fields include:

- `_id`
- `advisorId`
- `name`
- `channel`
- `subject`
- `body`
- `isActive`
- `createdAt`
- `updatedAt`

Templates must be persisted through MongoDB.

Requirements:

- Templates must belong to an existing AdvisorProfile.
- Template names and required content must be validated.
- Supported channel values must be enforced.
- Appropriate defaults must be applied.
- Inactive templates must not be used for new campaign activation.
- A Campaign must not use another Advisor's private Template.
- A Campaign's channel must match its selected Template's channel.

For example, an email campaign must not use a Facebook-specific template.

Existing Phase I template functionality must remain operational.

5.3 Replace the In-Memory Repository

Primary file: `backend/src/modules/campaigns/campaignRepository.js`

The existing repository currently stores Campaigns and Templates in temporary JavaScript Maps.

These operations must be replaced with asynchronous Mongoose queries.

Required repository functionality includes:

- Create Campaign.
- Find Campaign by ID.
- List Campaigns.
- Update Campaign.
- Create Template.
- Find Template by ID.
- List Templates.
- Filter Campaigns by advisor and status.
- Retrieve campaign targeting information.
- Update campaign specialist assignments.

Database queries must remain in the repository layer.

Campaign business rules must remain in the service layer.

Controllers must not directly query MongoDB.

The migration must preserve existing service interfaces wherever practical.

Campaign and Template records must remain available after server restart.

Filtering and pagination, where implemented, should occur through MongoDB queries rather than loading the entire collection into application memory.

5.4 Campaign Creation and Advisor Ownership

Existing endpoint:

`POST /api/v1/campaigns`

Campaign creation must require authentication.

An authenticated Advisor must be allowed to create Campaigns under their own AdvisorProfile.

The creation workflow must:

1. Authenticate the requesting user.
2. Verify the permitted role.
3. Resolve the authenticated Advisor's AdvisorProfile.
4. Determine the correct `advisorId`.
5. Validate the submitted Campaign information.
6. Validate any referenced Template.
7. Apply campaign defaults.
8. Save the Campaign to MongoDB.
9. Return the created Campaign.

If `advisorId` is supplied in the request body for compatibility, it must match the authenticated AdvisorProfile or be rejected.

A Marketing Specialist must not create a Campaign under an arbitrary Advisor ID.

For Phase II, general Campaign creation should remain an Advisor-owned operation. Specialists should manage Campaigns explicitly assigned to them.

Any future specialist-initiated Campaign creation must use a separately authorized workflow.

5.5 Campaign Retrieval and Authorization

Existing endpoints:

- `GET /api/v1/campaigns`
- `GET /api/v1/campaigns/:campaignId`

Campaign retrieval must require authentication.

The implementation must enforce the following:

Advisor access

An Advisor may retrieve Campaigns belonging to their AdvisorProfile.

An Advisor must not retrieve another Advisor's private Campaigns.

Specialist access

A Marketing Specialist may retrieve only Campaigns for which:

1. The specialist has an accepted relationship with the owning Advisor.
2. The Campaign contains the specialist's User ID in `assignedSpecialistIds`.

Administrator access

Administrative access must follow explicitly defined platform permissions.

Campaign listing must filter records according to the authenticated user's access rather than trusting arbitrary query parameters.

The backend must reject unauthorized resource access.

5.6 Campaign Updates and Lifecycle Enforcement

Existing endpoint:

`PATCH /api/v1/campaigns/:campaignId`

The existing Campaign lifecycle must be preserved.

Supported statuses:

- `DRAFT`
- `SCHEDULED`
- `ACTIVE`
- `COMPLETED`
- `CANCELLED`

The following transitions must remain supported according to the Phase I business rules.

Current Status	Allowed Next Status

DRAFT	DRAFT, SCHEDULED, ACTIVE, CANCELLED
SCHEDULED	DRAFT, SCHEDULED, ACTIVE, CANCELLED
ACTIVE	ACTIVE, COMPLETED, CANCELLED
COMPLETED	COMPLETED
CANCELLED	CANCELLED

Invalid transitions must be rejected.

Campaign status validation must be enforced by the backend.

Campaign activation requirements

Before activating a Campaign:

1. A valid active Template must exist.
2. The Template must belong to the Campaign's Advisor.
3. The Template channel must match the Campaign channel.
4. At least one eligible Lead must exist.
5. Marketing consent and unsubscribe conditions must be satisfied for email audiences.
6. Required Campaign fields must be valid.

A Campaign must not become `ACTIVE` when its activation requirements are not satisfied.

Scheduling

A Campaign entering `SCHEDULED` status must contain a valid future scheduling date.

The completed Campaign must retain its completion timestamp where applicable.

Specialist modification restrictions

Marketing Specialists may modify permitted content fields on assigned Campaigns.

For Phase II, the following restrictions are recommended:

- Specialists may edit titles, descriptions, targeting criteria, and permitted Template selections.
- Specialists must not modify `advisorId`.
- Specialists must not assign or remove other specialists.
- Specialists must not revoke advisor relationships.
- Specialists must not directly change Campaign status to `ACTIVE`, `COMPLETED`, or `CANCELLED` unless an explicit permission is defined.

Campaign lifecycle control should remain with the owning Advisor by default.

These restrictions must be enforced through the service layer.

5.7 Marketing Specialist Campaign Assignments

Campaign-specific specialist assignments are the responsibility of Member 3.

The invitation system itself belongs to Member 1.

The Campaigns module must use the shared advisor-specialist relationship contract.

Assignment requirements

An Advisor must be able to assign a Marketing Specialist to one of their own Campaigns.

Before an assignment is saved, the backend must verify:

1. The requesting user is the Campaign's owning Advisor or has an explicitly authorized administrative permission.
2. The Campaign exists.
3. The specialist exists.
4. The specialist has the `MARKETING_SPECIALIST` role.
5. An accepted advisor-specialist relationship exists.
6. The specialist is not already assigned to the Campaign.

The assignment must be stored persistently in the Campaign document.

Suggested endpoints

Method	Endpoint	Purpose

POST	`/api/v1/campaigns/:campaignId/specialists`	Assign specialist
GET	`/api/v1/campaigns/:campaignId/specialists`	List assigned specialists
DELETE	`/api/v1/campaigns/:campaignId/specialists/:userId`	Remove specialist

All endpoints must require authentication.

The Campaigns module must not implement a separate invitation system or duplicate the Identity module's assignment records.

Assignment removal

An Advisor must be able to remove a specialist from a Campaign.

After removal, the specialist must immediately lose authorization to access the Campaign.

If an advisor-specialist relationship is revoked, related Campaign permissions must also become invalid even if the specialist's User ID remains in the Campaign document.

Authorization must therefore verify the current accepted relationship, not merely the presence of a User ID.

Assignment API response

Assignment responses should identify assigned specialists without exposing private Identity information.

For example, an assignment list may contain:

- Specialist User ID.
- Name.
- Permitted public profile information.

Password hashes, credentials, and authentication secrets must never be returned.

5.8 Campaign Audience Selection

Existing endpoint:

`GET /api/v1/campaigns/:campaignId/audience`

Campaign audience selection must continue using Lead information from the Leads module.

Supported targeting criteria include:

- Lead interests.
- Lead lifecycle statuses.
- Advisor ownership.
- Marketing consent.
- Unsubscribe status.

The Campaign model may store audience criteria such as:

```
None
{
  "audience": {
    "interests": ["Retirement Planning"],
    "leadStatuses": ["NEW", "CONTACTED"]
  }
}
```

The audience selection workflow must:

1. Authenticate the requesting user.
2. Verify Campaign access.
3. Retrieve the Campaign.
4. Validate the Campaign's target audience criteria.
5. Retrieve matching Leads through the agreed Lead lookup contract.
6. Enforce Advisor ownership.
7. Exclude Leads without the necessary marketing consent.
8. Exclude unsubscribed Leads.
9. Return the eligible audience information.

The actual Lead database model and CRM business rules belong to Member 2.

Member 3 must not modify the Lead repository or duplicate the entire CRM implementation.

Specialist audience access

Marketing Specialists may need audience information to prepare campaigns.

However, they must not automatically receive unrestricted access to private CRM records.

For Phase II, Specialists should receive only the information necessary for their assigned Campaign operations.

Audience counts and non-sensitive targeting summaries may be exposed when permitted.

Full Lead contact records should remain restricted unless a separate explicit authorization policy is implemented.

The internal audience lookup contract must still support the Delivery module's server-side recipient eligibility checks.

5.9 Secure Template Access

Existing endpoints:

- `POST /api/v1/campaigns/templates`
- `GET /api/v1/campaigns/templates`
- `GET /api/v1/campaigns/templates/:templateId`

These endpoints must use MongoDB and shared authentication middleware.

Template ownership must be validated.

Advisors may manage their own Templates.

Marketing Specialists may access Templates required for their assigned Campaigns, subject to explicit permissions.

Specialists must not automatically be able to retrieve every Template belonging to an Advisor merely because one Campaign is assigned to them.

If a dedicated Template editing workflow is required, it must be restricted to the appropriate Advisor or authorized specialist.

The existing API behavior must be preserved wherever practical.

5.10 Secure Data Handling

The Campaigns module must implement:

- Authentication for protected endpoints.
- Role-based access enforcement.
- Advisor ownership checks.
- Specialist assignment verification.
- Validation of MongoDB ObjectIds.
- Validation of allowed request fields.
- Protection against mass assignment.
- Prevention of unauthorized ownership changes.
- Campaign lifecycle validation.
- Template ownership validation.
- Safe handling of database errors.
- Consistent HTTP error responses.

Client-supplied values must not be treated as proof of authorization.

Fields such as `advisorId`, `assignedSpecialistIds`, `createdAt`, and ownership-related metadata must not be freely modifiable through general Campaign update requests.

Specialist assignments must be managed through dedicated authorized endpoints.

Shared JWT validation and role middleware must be reused rather than reimplemented.

---

6. Internal Contracts for Other Modules

The Campaigns module must provide stable contracts for Delivery, Automation, Identity, and Leads.

These integrations must be supported without implementing other members' module responsibilities.

6.1 Identity Integration

Member 1 provides:

- JWT authentication.
- Role validation.
- AdvisorProfile lookup.
- Accepted advisor-specialist relationship lookup.

Member 3 must use these shared contracts to authorize Campaign operations.

Member 3 owns only the Campaign-specific assignment and access rules.

6.2 Leads Integration

Member 2 provides:

- Lead existence and lookup.
- Advisor ownership.
- Lead status.
- Lead interests.
- Marketing consent.
- Unsubscribe status.

Member 3 uses these attributes to determine Campaign audiences.

The Leads module must not be modified to accommodate private Campaign business logic unless an agreed contract change is necessary.

6.3 Delivery Integration

Member 4 must be able to retrieve and validate:

- Campaign existence.
- Owning Advisor ID.
- Campaign status.
- Campaign channel.
- Eligible audience information.
- Template-related information where necessary.

The Campaigns module must provide or preserve service interfaces for these operations.

For example, the Delivery module must be able to determine whether a Campaign is `ACTIVE` and whether its channel is `EMAIL` or `FACEBOOK`.

Member 3 must not implement delivery queues, actual message sending, delivery status tracking, or provider integrations.

These responsibilities belong to Member 4.

6.4 Automation Integration

Member 5 must be able to reference Campaigns through `campaignId`.

Campaign creation, status changes, and other relevant operations may produce business events.

Recommended event types may include:

- `CAMPAIGN_CREATED`
- `CAMPAIGN_ACTIVATED`
- `CAMPAIGN_COMPLETED`

Only event types established in the shared Automation contract should be emitted.

The event information must identify the related Campaign and authorized owner.

Event processing, ActivityEvent persistence, automation execution, and scheduling remain Member 5 responsibilities.

During independent development, event handling may be mocked or tested through an agreed interface.

The Campaigns module must remain functional without requiring the Automation module to be running during isolated tests.

---

7. API Compatibility

The following Phase I endpoints must remain functional:

Method	Endpoint	Purpose

POST	`/api/v1/campaigns`	Create Campaign
GET	`/api/v1/campaigns`	List Campaigns
GET	`/api/v1/campaigns/:campaignId`	Retrieve Campaign
PATCH	`/api/v1/campaigns/:campaignId`	Update Campaign
GET	`/api/v1/campaigns/:campaignId/audience`	Retrieve eligible audience
POST	`/api/v1/campaigns/templates`	Create Template
GET	`/api/v1/campaigns/templates`	List Templates
GET	`/api/v1/campaigns/templates/:templateId`	Retrieve Template

The following Phase II endpoints are recommended:

Method	Endpoint	Purpose
POST	`/api/v1/campaigns/:campaignId/specialists`	Assign Marketing Specialist
GET	`/api/v1/campaigns/:campaignId/specialists`	List Campaign specialists
DELETE	`/api/v1/campaigns/:campaignId/specialists/:userId`	Remove Marketing Specialist

The existing route paths and response formats should be preserved wherever practical.

Phase II endpoints must be protected according to the established role and resource permissions.

Additional API changes must be documented.

---

8. Testing Requirements

The Campaigns module must be tested using Postman and appropriate automated tests.

Tests must use MongoDB persistence, realistic sample data, and controlled test fixtures.

At minimum, the following scenarios must be verified.

8.1 Database Persistence

- Campaign creation stores records in MongoDB.
- Template creation stores records in MongoDB.
- Campaign updates remain persistent.
- Template records remain available after restart.
- Campaign references are stored correctly.
- Invalid schemas are rejected.
- Database indexes and validation rules are applied.

8.2 Advisor Ownership

- Advisor can create their own Campaign.
- Advisor can retrieve their own Campaigns.
- Advisor can modify their own Campaign.
- Advisor cannot access another Advisor's private Campaign.
- Advisor cannot modify another Advisor's Campaign.
- Advisor cannot change Campaign ownership through an ordinary update.
- Supplied `advisorId` cannot override authenticated ownership.

8.3 Authentication and Role Enforcement

- Missing JWT is rejected.
- Invalid JWT is rejected.
- Expired JWT is rejected.
- Unauthorized roles are rejected.
- Unassigned specialists cannot access private Campaigns.
- Specialists assigned to unrelated Advisors cannot access Campaigns.
- Authorized Advisors and assigned Specialists can perform permitted operations.

8.4 Template Validation

- Template creation succeeds with valid information.
- Invalid Template data is rejected.
- Inactive Templates cannot be used for activation.
- Templates belonging to another Advisor are rejected.
- Campaign and Template channels must match.
- Missing Template references are handled correctly.

8.5 Campaign Lifecycle

- Campaigns can be created in `DRAFT` status.
- Valid status transitions succeed.
- Invalid status transitions are rejected.
- Scheduled Campaigns require a future date.
- Activation requires a valid Template.
- Activation requires an eligible audience.
- Completed Campaigns cannot be improperly reactivated.
- Cancelled Campaigns cannot be improperly reactivated.

8.6 Specialist Assignments

- Advisor can assign a specialist with an accepted relationship.
- Advisor cannot assign a specialist without an accepted relationship.
- Advisor cannot assign a User with an invalid role.
- Advisor cannot assign a specialist to another Advisor's Campaign.
- Duplicate specialist assignments are rejected or handled idempotently according to the documented contract.
- Assigned specialists can access permitted Campaign resources.
- Unassigned specialists are denied.
- Specialists cannot assign themselves.
- Specialists cannot remove other specialists.
- Advisor can remove a specialist.
- Removed specialists immediately lose Campaign access.
- Revoked advisor-specialist relationships invalidate Campaign access.

8.7 Audience Selection

- Eligible Leads are selected correctly.

- Lead interest filtering works.
- Lead status filtering works.
- Advisor ownership is enforced.
- Leads without marketing consent are excluded.
- Unsubscribed Leads are excluded.
- Campaign activation fails when no eligible audience exists.
- Specialists cannot retrieve unrestricted private Lead information.
- Unauthorized audience requests are rejected.

8.8 Integration Contract Tests

- Campaigns reference valid AdvisorProfiles.
- Campaign Templates reference valid owners.
- Campaigns maintain correct specialist User references.
- Delivery can retrieve Campaign status and channel.
- Delivery can validate Campaign audience eligibility.
- Automation can reference Campaign IDs through agreed event contracts.
- Campaign operations remain functional when other feature modules are mocked.

---

9. Documentation and GitHub Requirements

Member 3 must provide module-specific documentation explaining:

- Campaigns and Templates module responsibilities.
- MongoDB Campaign schema.
- MongoDB Template schema.
- Field definitions and validation rules.
- AdvisorProfile relationships.
- Template relationships.
- Specialist assignment relationships.
- Campaign lifecycle statuses and transitions.
- Campaign activation rules.
- Audience selection logic.
- Marketing consent considerations.
- Role-based authorization.
- Specialist permissions and restrictions.

- API endpoint usage.
- Internal integration contracts.
- Postman test scenarios.
- Any necessary changes from Phase I.

The module documentation must explicitly state:

`advisorId` references `AdvisorProfile._id`, not `User._id`.

`assignedSpecialistIds` references `User._id` records with the `MARKETING_SPECIALIST` role.

An accepted advisor-specialist relationship alone does not authorize access to every Campaign.

No database credentials, JWT secrets, or sensitive user information must be committed to GitHub.

Development must occur in:

`feature/campaign-p2-m3`

Changes must be committed with meaningful messages.

A pull request must be submitted targeting `main`.

The pull request must include:

- Completed functionality summary.
- Schema and repository changes.
- Authentication and authorization integration.
- Campaign assignment implementation.
- Tests performed.
- Relevant setup instructions.
- Known limitations, if applicable.

The module must remain independently testable before merging.

---

10. Completion Criteria

Member 3's Phase II responsibilities will be considered complete when:

- Campaign and Template models are integrated with MongoDB using Mongoose.
- In-memory Campaign and Template storage has been replaced.
- Campaign and Template records remain available after restart.
- Campaigns and Templates reference valid AdvisorProfiles.
- `advisorId` consistently means `AdvisorProfile._id`.
- Template references and channel compatibility are validated.
- Required fields, defaults, indexes, and schema validation are implemented.
- Existing Campaign and Template endpoints remain functional.
- Campaign and Template endpoints use JWT authentication.
- Advisor ownership is enforced by the backend.
- Marketing Specialists can access only permitted assigned Campaigns.
- Campaign-specific specialist assignments are persistent.
- Specialist assignments require accepted advisor relationships.
- Advisors can add and remove specialists from their own Campaigns.
- Revoked relationships invalidate specialist Campaign access.
- Campaign lifecycle transitions are enforced.
- Campaign activation requirements are validated.
- Audience selection uses eligible Lead records.
- Non-consenting and unsubscribed Leads are excluded from email audiences.
- Unauthorized access to private Lead information is prevented.
- Delivery can retrieve required Campaign information through stable contracts.
- Automation event references follow the agreed contracts.
- No unrelated module's business logic has been implemented unnecessarily.
- Postman and automated tests cover successful and unsuccessful operations.
- Module documentation is complete.
- Changes are committed and submitted through a pull request.

Expected outcome: The Marketing Campaigns and Templates module will operate using persistent MongoDB storage, with secure Advisor ownership, enforced Campaign lifecycle rules, consent-aware audience selection, and campaign-specific Marketing Specialist assignments. Advisors will maintain ownership of their Campaigns, while authorized Marketing Specialists will be able to work only on resources explicitly assigned to them. The module will support Delivery and Automation through stable backend contracts without taking responsibility for those modules' implementations.